

Indian Visa Portal

Website Redesign

Learn the Importance of Menus and Navigation

Subject	Human Machine Interaction
Class	T.E. Computer Engineering — Sem V
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Department	Computer Engineering, CRCE

Project Team

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GitHub Repository

Department of Computer Engineering

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Title	Learn the importance of menus and navigation
Date of Performance	
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Roll No.	10584, 10613, 10616, 10617
Name of the Students	Jeet Chavan, Sanika Lobo, Meet Mangaonkar, Yash Masaye

Sr. No	Rubric	Grade
1	Analysis (2)	
2	Improvement (2)	
3	HMI design principles (2)	
4	Test Cases and conclusion (2)	
5	Post Lab Assignments (2)	
	Total Marks awarded: (10)	

Signature of the Teacher: _____

Aim

To redesign an existing cluttered and poorly structured website — the Indian Visa application portal — so that it is cleaner, more navigable, contextually intelligent, and adheres to HMI design principles.

Theory

News and civic portals like the original Indian Visa website present a classic HMI problem: information overload without prioritisation. The user must spend considerable time finding even basic information, and the portal makes no attempt to tailor its content to the user's context, location, or intent.

The core challenge is to apply principles of HMI design — particularly **cognitive load reduction, progressive disclosure, and contextual relevance** — to transform a portal that works as a dumping ground for content into one that works as an intelligent guide for the applicant. Just as a person sitting in Russia should not receive US-centric top news as a priority, a first-time tourist applicant should not have to wade through e-Visa policy amendments for researchers in order to find the “Apply” button.

Task 1 — Analysis of Existing Portal (indianvisaonline.gov.in)

The existing Indian Visa Online portal was evaluated for usability, navigation clarity, and HMI compliance. The following problems were identified:

1. Visual Clutter & Information Overload

The homepage presents dense blocks of text, multiple scrolling marquee announcements, and a navigation bar with too many ungrouped items presented simultaneously. Users cannot identify a clear entry point for their specific task.

2. Poor Navigation Hierarchy

Menus are flat with no logical grouping. There is no distinction between primary actions (Apply, Track) and secondary content (Advisories, FAQs), violating Fitts's Law and the principle of visual hierarchy.

3. No Geo-contextual Personalisation

The portal presents identical content to all users regardless of nationality, location, or visa type interest. A UAE citizen applying for a business visa and a US tourist see the same undifferentiated homepage.

4. No Responsive / Mobile Design

The legacy portal was not designed for mobile viewports. Tables and form elements overflow on small screens, making the experience nearly unusable on smartphones — which constitute a significant share of the user base.

5. Accessibility Gaps

The portal lacks semantic HTML, skip-navigation links, proper ARIA attributes, and keyboard navigation support — failing WCAG 2.1 basic compliance for a government-facing website.

6. No Intelligent Search or Discovery

There is no Visa Finder tool, no guided wizard, and no way to narrow information by purpose of travel. Users must already know their visa category, placing excessive cognitive burden on them.

7. Single Language Only

The portal is English-only, excluding millions of applicants from non-English-speaking countries who may struggle with legal or procedural language without a translation option.

Task 2 & 3 — Redesigned Solution: Indian Visa Modern Portal

The redesigned portal was built as a full React + TypeScript single-page application. The key improvements are described below.

Key Features Implemented

Feature	Description
Visa Finder Wizard	An interactive, step-by-step tool that asks the user their nationality and travel purpose, then narrows down the correct visa category — replacing cognitive burden with guided discovery.
Simplified Application Flow	A clearly sectioned multi-step form with a progress indicator. Each step presents only what is required at that moment, implementing progressive disclosure.
Application Tracking	A dedicated tracking page where applicants enter their reference number to retrieve a clear status update — promoted to top-level navigation as a primary action.
Plan Journey	Interactive map and destination guides built with React Simple Maps / D3-Geo, helping users contextualise their trip within the same portal.
Travel Advisories	Advisories are separated into their own section, clearly labelled, and sorted by urgency — not mixed into the homepage alongside application instructions.
Multi-language Support	A custom <code>useLanguage</code> hook and <code>/i18n</code> locale directory allow seamless, no-reload language switching for applicants globally.
Accessibility First	A dedicated <code>AccessibilityProvider</code> manages high-contrast and reduced-motion preferences. Skip-to-content links and ARIA attributes are included throughout.
Fully Responsive	Built mobile-first with semantic HTML, flexbox and grid layouts — fully functional across all screen sizes and devices.

Tech Stack

Technology	Version	Role
React	19.2	UI component library and rendering framework
TypeScript	~6.0	Strict type safety for scalable, maintainable code
Vite	8.3	Fast development server and optimised production build
React Router	7.18	Client-side routing across 10+ pages
Lucide React	1.47	Consistent, accessible icon system
React Simple Maps / D3-Geo	latest	Interactive geographic map for journey planning
Oxlint	1.81	Fast JS/TS linting for code quality assurance

HMI Design Principles Applied

1. Visibility of System Status

The multi-step application form always shows the user where they are in the process via a progress bar. The Track page displays the precise status of their submission clearly.

2. Match Between System and Real World

Navigation labels use natural language (“Apply,” “Track,” “Help”) rather than government jargon. The Visa Finder asks questions the user naturally thinks in (“Where are you from?”, “Why are you visiting?”).

3. User Control and Freedom

Every multi-step form has a clear “Back” option. Users can switch language at any point without losing state. Account management allows reviewing and editing past submissions.

4. Consistency and Standards

Lucide icons and a consistent design token system (colour, spacing, typography) are used throughout all 10+ pages, reducing learning curve and confusion.

5. Reduction of Cognitive Load

Navigation groups primary tasks (Apply, Track, Find) separately from informational content (Advisories, Help, Plan). The Visa Finder replaces free-text search with guided choices, reducing burden on working memory.

6. Flexibility and Efficiency of Use

A global Search page allows power users to jump directly to any section. Keyboard navigation and skip-to-content links serve users who avoid the mouse.

7. Aesthetic and Minimalist Design

Each page shows only what is relevant to the user's current task. Advisories, journey planning, and application steps are cleanly separated. The homepage centres on a single primary call-to-action.

Test Cases

The redesigned portal was demonstrated to five classmates who provided feedback. Tests covered the five most common user journeys.

#	Scenario	Expected Outcome	Observed Result	Status	
1	New user applying for tourist visa	Reach application form in ≤ 2 clicks via Finder or Apply button	All 5 testers reached form in ≤ 2 clicks; no confusion about entry point	PASS	
2	Track existing application by reference number	Track page discoverable from nav; status shown clearly	"Track" found in top nav by all testers without guidance	PASS	
3	Switch portal language without page reload	Language switcher accessible from any page; content updates instantly	4/5 testers found it; 1 initially missed it in the header	PARTIAL	
4	Find active travel advisories	Advisories on dedicated page, sorted by urgency	All testers located Advisories; content clearly separated from application steps	PASS	
5	Use Visa Finder to identify correct visa type	Wizard narrows to correct category within 3 steps	Wizard logic correct; testers preferred it over legacy site's flat list	PASS	

Results and Conclusion

The redesigned Indian Visa portal successfully addresses all major usability problems identified in the analysis phase. By applying core HMI principles — reducing cognitive load, establishing clear navigation hierarchy, implementing progressive disclosure, and ensuring accessibility — the portal is transformed from a confusing information dump into a task-focused, guided experience.

All five classmate testers completed primary tasks faster and with less confusion compared to the original portal. Test Case 3 (language switch discoverability) yielded a partial pass, indicating that the language control could be made more visually

prominent — a clear direction for future iteration.

The use of React, TypeScript, and Vite ensures the redesign is not only functionally superior but also maintainable, type-safe, and performant — a solid foundation for a real-world civic application.

Post Lab Assignment Questions

Q1. List and explain problems with the existing web site.

The existing Indian Visa Online portal suffers from: (i) visual clutter with no clear hierarchy on the homepage; (ii) a flat, ungrouped navigation menu that conflates primary actions with secondary content; (iii) no geo-contextual personalisation — every user sees identical content regardless of nationality or intent; (iv) no responsive design, making it nearly unusable on mobile; (v) missing accessibility features such as skip links, ARIA attributes, and keyboard navigation; (vi) no guided Visa Finder — users must already know their visa category; and (vii) English-only content excluding non-English-speaking applicants.

Q2. How does your design show improvement over the existing design?

The redesigned portal improves on every identified problem: a clean, task-focused homepage with a single primary CTA replaces the cluttered original; navigation is grouped into primary tasks (Apply, Track, Find) and secondary content (Advisories, Help, Plan); a Visa Finder wizard introduces guided contextual discovery; the layout is fully mobile-responsive; a dedicated **AccessibilityProvider** handles high-contrast and reduced-motion preferences with skip-to-content links and semantic HTML; multi-language support via a custom **i18n** system is available site-wide; and the application flow uses progressive disclosure (multi-step form with progress indicator) rather than presenting all fields at once.

Q3. Which software / tool have you used to design your application?

The application was designed and developed using **React 19.2** with **TypeScript (~6.0)** as the UI framework, **Vite 8.3** as the build tool and development server, **React Router 7.18** for client-side routing, **Lucide React** for iconography, **React Simple Maps with D3-Geo** for the interactive journey planning map, and **Oxlint** for code quality assurance. Version control was managed with **Git** and the project is hosted on **GitHub** at **Jeetchavan02/Indian-Visa**.